A Tour of Multiple View Geometry: review

Multiple View Geometry in Computer Vision, chapter 1

Lesson t3.1 · Hartley, Zisserman

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What is on these slides

1. Key result per section, with its trap
2. Cards C1–C9: the code that matters
3. Shapes and cost per card
4. Symptom to fix and mistakes already made
5. Conventions and where claims are proved

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1.1 Homogeneous coordinates, X′ = HX

$$(kx,\ ky,\ k)^\top \sim (x,\ y,\ 1)^\top$$ p. 2

$$\mathbf{X}' = \mathrm{H}_{(n+1)\times(n+1)}\mathbf{X}$$ p. 3

k non-zero scale

(x, y, 0)^T point at infinity

H non-singular, up to scale

- Points at infinity have last coordinate 0; H is 8 DoF (3×3), 15 DoF (4×4)
- Trap: dehomogenizing a point at infinity; test |w| against ‖x‖, raise
- Print slip p. 3: line and plane at infinity swapped with space and image

Dehomogenizing infinity is the trap: test |w|, then raise

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1.1.1 Ladder: Ω∞ and perpendicularity

$$\mathbf{d}_1^{\top}\,\Omega_{\infty}\,\mathbf{d}_2 = 0$$

p. 6

Ω∞ absolute conic, T = 0

$\mathbf{d}_1, \mathbf{d}_2$ direction vectors

- Circular points $(1, \pm i, 0)^{\top}$ lie on every circle (p. 5); Ω∞ = I in a Euclidean frame
- Affine: line at infinity fixed; similarity: circular points fixed too
- Trap: affine corner 49.0° and opposite sides 0.0° apart; right angles lost

Affine maps lose right angles even though parallels survive

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1.2 Camera and plane homography

$$(x,\ y,\ w)^\top = \mathrm{P}_{3\times 4}\,(X,\ Y,\ Z,\ T)^\top \qquad \text{p. 7}$$

$$H = [\,\mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_4\,] \quad (Z = 0) \qquad \text{p. 7}$$

P camera, 3×4, rank 3

H plane homography, 3×3

(X, Y, Z, T) world point

- Rank 3, centre PC = 0 (§6.2.1, p. 158); same centre means H, any scene
- Trap: centre on the plane makes H rank 2; depth 4–12 m breaks H at 0.048 m

Centre on the plane makes H rank 2; depth breaks H at 0.048 m

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1.2 Calibrated: angle between rays

$$\cos\theta = \frac{\mathbf{x}_1^\top \omega \mathbf{x}_2}{\sqrt{(\mathbf{x}_1^\top \omega \mathbf{x}_1)(\mathbf{x}_2^\top \omega \mathbf{x}_2)}} \tag{8.9}$$

$\mathbf{x_1}, \mathbf{x_2}$ homogeneous pixels

ω $(KK^T)^{-1}$, 3×3

θ ray angle

- pp. 9–10; $\omega = (KK^T)^{-1}$ (Result 8.17, p. 210); Example 1.1 is the ladder from one image
- Trap: f 20% short reads 6.04° high at the edge; use atan2, not arccos

A short f reads high at the edge; use atan2, not arccos

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1.3 Ambiguity and F

$$P_j \mathbf{X}_i = (P_j H^{-1})(H \mathbf{X}_i) \tag{1.1}$$

$$\mathbf{x}_i'^{\top} F \mathbf{x}_i = 0$$

p. 11

P_j camera j

X_i scene point

H any 4×4

F rank 2, 3×3

- pp. 10–12; rank F = 2 (det F = 0, p. 13); the outline is not for implementing
- Trap: compare reprojected pixels, not P and P H^{-1}; measure $x'^{T}Fx$ in pixels

Compare reprojected pixels, not P and P H^{-1}

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1.7 Euclidean upgrade

$$\Omega_\infty = M^\top \omega M = L L^\top$$ §10.4

Ω∞ absolute conic

M camera left block

L Cholesky factor

- pp. 16–17: Ω∞ lies on π∞; two views leave two solutions, three leave one
- Trap: ω has arbitrary sign; π_4 near 0 needs a Householder step

ω has arbitrary sign; fix it before the Cholesky step

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1.4–1.8 Skims in one place

- 1.4 Trifocal tensor T_i^{jk}: 27 entries, 4 equations per triplet, (1.2) p. 12
- 1.5 Affine factorization by SVD; bundle adjustment needs a start (p. 14)
- 1.6 Transfer $x'' = P''X$ (p. 16); plane or pure rotation: one view, H
- 1.8 Auto-calibration: π_∞ hard, ω linear; square pixels, 4 images

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C1 Homogeneous points: scale-free infinity test

```python
def normalize_h(Xh):
    Xh = np.atleast_2d(np.asarray(Xh, float))
    return Xh / np.linalg.norm(Xh, axis=1, keepdims=True)

def at_infinity(Xh, tol=1e-12):
    return np.abs(normalize_h(Xh)[:, -1]) < tol

def dehom(Xh, tol=1e-12):
    Xh = np.atleast_2d(np.asarray(Xh, float))
    if np.any(at_infinity(Xh, tol)):
        raise ValueError("point at infinity")
    return Xh[:, :-1] / Xh[:, -1:]
```

- Rows are points; dehomogenize last and raise at infinity

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C3 Camera centre by SVD, depth by eq. 6.15

```python
def camera_centre(P, rank_tol=1e-12):
    _, s, Vt = np.linalg.svd(P)
    if s[2] < rank_tol * s[0]:
        raise ValueError("rank P < 3: not a camera (p. 7)")
    return Vt[-1]

def depth(P, Xh):
    Xh = np.atleast_2d(Xh)
    M = P[:, :3]
    w = Xh @ P[2]
    return (np.sign(np.linalg.det(M)) * w
            / (Xh[:, 3] * np.linalg.norm(M[2])))
```

- Negative depth: behind the camera, same pixel as a point in front

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C4 Plane homography and the pixel fit test

```python
def plane_homography(P, cond_max=1e12):
    H = P[:, [0, 1, 3]]
    if np.linalg.cond(H) > cond_max:
        raise ValueError("plane Z = 0 through the centre")
    return H / np.linalg.norm(H)

def transfer_rms_px(H, x1, x2):
    y = np.atleast_2d(x1) @ H.T
    d = y[:, :2] / y[:, 2:] - x2[:, :2] / x2[:, 2:]
    return np.sqrt(np.mean(np.sum(d ** 2, axis=1)))
```

- Pick columns, do not fit; normalize by Frobenius norm, not H_{33}

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C5 Projective ambiguity: solve, do not invert

```python
def apply_ambiguity(Ps, Xh, H, cond_max=1e10):
    if np.linalg.cond(H) > cond_max:
        raise ValueError("H is numerically singular")
    Ps_new = [np.linalg.solve(H.T, P.T).T for P in Ps]
    Ps_new = [P / np.linalg.norm(P) for P in Ps_new]
    Xh_new = np.atleast_2d(Xh) @ H.T
    Xh_new /= np.linalg.norm(Xh_new, axis=1, keepdims=True)
    return Ps_new, Xh_new
```

- $P\,H^{-1} = (H^{-T}P^{T})^{T}$: one LU of H^{T}; the oracle is the pixel change

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C6 Epipolar distance in pixels, scale-free rank

```python
def epipolar_distance_px(F, x, xp):
    x, xp = np.atleast_2d(x), np.atleast_2d(xp)
    l = x @ F.T
    return (np.abs(np.sum(xp * l, axis=1))
            / np.hypot(l[:, 0], l[:, 1]))

def rank2_gap(F):
    s = np.linalg.svd(F, compute_uv=False)
    return s[2] / s[0]
```

- Normalize points before step (i); same centre gives F = 0

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C7 Linear triangulation, any number of views

```python
def triangulate(Ps, xs):
    rows = []
    for P, x in zip(Ps, xs):
        x = x / x[2]
        rows += [x[0] * P[2] - P[0], x[1] * P[2] - P[1]]
    A = np.array(rows)
    A /= np.linalg.norm(A, axis=1, keepdims=True)
    X = np.linalg.svd(A)[2][-1]
    ...  # unit-norm X: see translate.md C7
```

- Homogeneous DLT, row scaling; check cheirality and reprojection

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C8 Ray angle: back-project, then atan2

```python
def back_project(K, x):
    return solve_triangular(K, np.atleast_2d(x).T,
                            lower=False).T

def ray_angle(K, x1, x2):
    d1, d2 = back_project(K, x1), back_project(K, x2)
    cross = np.linalg.norm(np.cross(d1, d2), axis=1)
    dot = np.sum(d1 * d2, axis=1)
    ...  # atan2 of cross and dot: see translate.md C8
```

- No inverse of KK^T; arccos loses about half the digits near zero

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C9 Metric upgrade by one Cholesky

```python
def metric_upgrade(M, omega):
    Omega = M.T @ omega @ M
    Omega = 0.5 * (Omega + Omega.T)
    if np.trace(Omega) < 0:
        Omega = -Omega
    ...  # Cholesky of Omega: see translate.md C9
    H = np.eye(4)
    H[:3, :3] = L.T
    return H
```

- Equals Result 10.5's H up to a rotation: $A = L^{-T}Q$
- Upgrade cameras and points together with C5

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Naive or robust, card by card

Naive

- Divide by w, get infinity
- arccos for the ray angle
- Compare P with $P\,H^{-1}$
- Invert H and KK^{T}

Robust

- Test |w| against ‖x‖, then raise
- atan2 of cross and dot
- Compare reprojected pixels
- Solve with one LU; triangular solves

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Each robust column entry is one of the cards above

Session readouts at default settings

0.140

px, distance of x′ to its epipolar line

0.740

px, median over all 60 points

6.90

px, rotation-only H on floor points

5.55

px, rotation-only H on clutter

- Python session and JS session agree on all 15 settings

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Shapes and cost per card

Card	Shapes	Cost
C1	$x \in \mathbb{R}^3$, $X \in \mathbb{R}^4$, H $(n+1)^2$	9 / 16 multiplies per point
C3	P 3×4, rank 3	12 multiplies per point
C4	H 3×3	9 multiplies per point
C5	H 4×4	one 4×4 LU; 48 per camera
C6	F 3×3, rank 2	one 3×4 SVD for F
C7	A is 2n×4	one 2n×4 SVD per point
C8	K, ω 3×3	two triangular solves
C9	Ω∞ 3×3	O(1) per upgrade

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Symptom to fix

Symptom	Cause	Try
F flips between frames	Rotation or planar scene	Fit H and F, pick by score
Parallels converge	π_∞ unknown	Calibrate K; impose parallels
Right angles wrong	Ω_∞ (ω, K) wrong	Recalibrate f, aspect, skew
Bearings off at the edge	Wrong f in ω	Recalibrate
Map size drifts	Scale unobservable	IMU, stereo, scale prior
BA diverges	Bad start or free gauge	Initialize; anchor pose and scale
Panorama ghosting	Centre moved	Rotate about the centre
Bird's-eye warps	Stale horizon	Update H from attitude

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Mistakes already made (REVIEW.md)

- PC = 0 is §6.2.1, p. 158; p. 7 gives only rank 3 and the [I | 0] centre
- Cheirality (which side is in front) is ch. 21, p. 515, not p. 11
- Seven matches fix F is a dof count; the book gives no reason (p. 281)
- The vanishing point is finite; the ideal point is the wall direction
- H_M differs from Result 10.5's H by a rotation: $A = L^{-T}Q$
- Example 1.1 is pp. 9–10, not p. 10

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The one sentence

Images fix a scene only up to a group; each extra piece of knowledge removes part of it

Projective, then affine with π_∞, then metric with Ω_∞; scale stays free

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Conventions that bite

- P is a 3×4 camera, not a covariance; H is a homography, not a Jacobian
- H&Z P = K[R | t] maps world to camera; GTSAM Pose3 is the inverse
- ω is the image of the absolute conic, not an angular rate
- T is the fourth homogeneous coordinate
- Noise, outliers and RANSAC are not in this chapter (§4.7)

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Where each claim is proved

- Ch. 2–3: lines, conics, Ω∞, angle formula (3.23)
- Ch. 6–8: K[R | t], IAC; ch. 9 and 11: F; ch. 12: triangulation
- Ch. 18–19: factorization, bundle adjustment, auto-calibration
- Robot side: lesson t2.8 graph SLAM; scale from the IMU

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